

Course Content		
Unit	Topic	No. of Teaching Hours
1.	Operating Systems: Overview, Purpose & Functionality, Evolution, Needs, Principles, Computing Environments; Device Organization & System Operations: Interrupts, Dual Mode Execution, Single & Multiuser; Services & System Calls, Shell Management; and OS Structuring Techniques & Design Issues.	5
2.	Process Management: Concepts, States, Structures, Context Switching; Operations on Processes; IPC Issues; Threads: Parallelism & Concurrency. Scheduling & Dispatching: Types, and Algorithms;	8
3.	Process Synchronization: Primitives, Critical Section Problem & Solutions. Deadlocks: Characterization, and Handling.	6
4.	Memory Management: Physical Memory & Issues; Allocation Techniques: Contiguous & Non-Contiguous; Virtual Memory: Demand Paging, Working Sets, and Thrashing.	5
5.	Storage Management: Disk Management, Disk Scheduling, Disk Structure, Swap-Space Management, and File System & Implementation.	4
6.	Security & Protection: Overview, Significance, Policy/Mechanism Separation, Security Methods & Devices Protection, and Access Control & Authentication.	2
Total Contact Hours		30